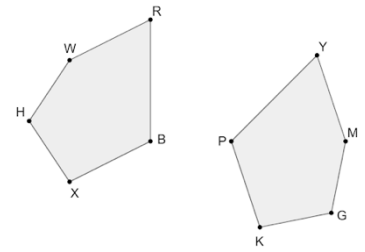
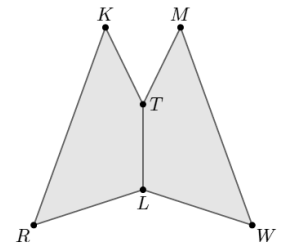


Geometry: Unit 5, Lesson 10

1. Polygons $RBXHW$ and $YPKGM$ are shown, where $YM = 5x$, $RB = 7x - 3$, $HW = 36$, $PK = YM$, and $HX = HW$. Polygon $RBXHW$ is congruent to polygon $YPKGM$. The perimeter of polygon $YPKGM$ is 222 units. What equation can be used to determine the value of x ?



2. Anna is making a pattern for a dog outfit. One part of the pattern is shown, where $KTLR \cong MTLW$ and $m\angle TLR = 108^\circ$, $m\angle WMT = 46^\circ$, and $m\angle MWL = 52^\circ$. What is the measure of $\angle LTK$?



3. A landscaping engineer used computer software to draw a garden she is designing. Her drawing is shown. Line segments DL , RB , TF , GR , FH , and GH represent paths around and through the garden. Quadrilaterals $MWGR$ and $KWHF$ are congruent, and triangles HMW and GKW are congruent where $WG = 35$ meters (m), $RG = 51.3m$, $WM = 38m$ and $RM = 40m$. What is the total distance someone will walk if they enter at point K and walk along the paths made by the quadrilateral $KWHF$?

